

chodroff2017

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1 Dataset overview

The dataset accompanies Chodroff & Wilson’s work on the structure of laryngeal contrast and voice-onset time in American English. Twenty-eight native English speakers produced monosyllabic words beginning with one of the six English stops (/p t k b d g/). For every stop token the recordings were force-aligned and a Praat script measured the stop interval, the duration of the following sonorant, the containing word, and a ten-point fundamental frequency contour sampled every 5 ms from vowel onset.

1.1 Measures analysed

We contrast three acoustic measures across talker gender:

- **VOT** (ms) — the stop’s voice-onset time, i.e. the interval from burst release to voicing onset, the standard voicing cue for stops. In this corpus the annotated stop interval *is* the VOT, so VOT and the stop’s own segment duration are the same quantity here.
- **Vowel duration** (ms) — the length of the vowel that follows the stop, taken from the following-sonorant duration. As the README notes, once the consonant-cluster (“L”) tokens are removed the following sonorant is always the vowel.
- **F0** (Hz) — the fundamental frequency at vowel onset, summarised as the mean of the first five 5-ms samples (the first 25 ms of the vowel), following the corpus convention. F0 largely reflects vocal-fold anatomy, so it is included partly as a reference measure (see the interpretation).

VOT is a property of the consonant’s laryngeal timing, vowel duration reflects the length of the following vowel, and F0 reflects vocal-fold vibration rate. Comparing all three shows whether a gender difference lives in the consonant cue, the vowel, the pitch, or some combination.

1.2 Preprocessing

The raw data is preprocessed following the steps below:

- Select the target stops, i.e. /t/, /d/;
- Remove consonant-cluster tokens. Cluster words such as *KLOT* or *PLOT* (CLVC) are flagged by a following sonorant of "L"; these are dropped so that the segment following every retained stop is a vowel and all three measures are comparable.
- VOT and the following-vowel duration are stored in seconds and re-expressed in milliseconds.
- The ten F0 samples are stored as text with "--undefined--" where Praat could not track pitch. These frames are read as missing, as are physiologically implausible values (outside 50–400 Hz, which are octave-tracking artefacts). F0 is then summarised as the mean of the first five samples (first 25 ms of the vowel); the few tokens left with no usable F0 sample are dropped from the F0 comparison only.

From the 8994 raw stop tokens, restricting to /t/ and /d/ and dropping 0 consonant-cluster tokens leaves 2201 tokens from 24 talkers (109 of which lack a usable F0 sample and are excluded from the F0 comparison only). Their distribution across stop and talker gender is shown in Table 1.

Table 1: Number of /t/ and /d/ tokens and contributing talkers, by talker gender, in the final analysis set.

Stop	Gender	Tokens	Talkers
T	female	575	13
T	male	473	11
D	female	628	13
D	male	525	11

2 Analyze phoneme duration, VOT, and F0

2.1 Goal and motivation

The goal is to compare three acoustic measures — the stop’s **VOT**, the following **vowel duration**, and the vowel-onset **F0** — between female and male talkers, separately for /t/ and /d/. VOT indexes the laryngeal timing of the consonant, vowel duration indexes the length of the following vowel, and F0 indexes the rate of vocal-fold vibration. Examining all three lets us see which properties carry a gender difference. F0 is expected to differ substantially for anatomical reasons and so serves as much as a reference (a “does the gender split behave as known?” check) as a finding in its own right.

The comparison is made on **per-talker means rather than raw tokens**. Each talker contributes many tokens, and tokens from the same talker are not independent; collapsing to one mean per talker (for each stop and measure) yields a single independent observation per talker, the appropriate unit

for a between-groups test. We then compare the female and male talker means with a two-sided **Welch two-sample t-test** (unequal-variance form), run once for each of the six stop $\times$ measure combinations.

2.2 Results

Table 2: Female vs. male comparison of VOT, vowel duration, and F0 for /t/ and /d/, on per-talker means. Columns give the female and male group means, their difference (female - male), the Welch t statistic, degrees of freedom, p-value, and the 95% CI on the difference. Units follow the Measure column: ms for VOT and vowel duration, Hz for F0.

Measure	Stop	Female	Male	Diff	t	df	p	95% CI
F0 (Hz)	/t/	218.7	126.5	92.2	10.50	20.7	< .001	[73.9, 110.5]
F0 (Hz)	/d/	201.4	118.7	82.7	10.07	17.4	< .001	[65.4, 100.0]
VOT (ms)	/t/	107.2	86.3	20.9	2.44	21.5	0.023	[3.1, 38.7]
VOT (ms)	/d/	22.5	18.6	4.0	2.37	21.9	0.027	[0.5, 7.4]
Vowel duration (ms)	/t/	151.8	159.4	-7.7	-0.69	20.5	0.496	[-30.7, 15.4]
Vowel duration (ms)	/d/	184.7	181.3	3.4	0.31	22.0	0.757	[-19.3, 26.2]

The three measures behave differently. **VOT** shows a significant female-longer difference for both stops: /t/ 107.2 vs. 86.3 ms ($t(21.5) = 2.44$, $p = 0.023$); /d/ 22.5 vs. 18.6 ms ($t(21.9) = 2.37$, $p = 0.027$). **Vowel duration** shows *no* reliable gender difference for either stop (/t/ $t(20.5) = -0.69$, $p = 0.496$; /d/ $t(22.0) = 0.31$, $p = 0.757$), with confidence intervals spanning zero. **F0**, as expected, is much higher for female talkers — /t/ 219 vs. 127 Hz ($t(20.7) = 10.50$, $p < .001$); /d/ 201 vs. 119 Hz ($t(17.4) = 10.07$, $p < .001$) — a difference of roughly $1.73\times$ in pitch. The full set of estimates and tests is in Table 2, and the underlying per-talker distributions are shown in Figure 1.

2.3 Interpretation

Bringing the three measures together sharpens the picture. The gender difference is **carried by the consonant’s VOT and by F0, but not by vowel duration**. Female talkers have reliably longer VOT for both stops (about 20 ms for the aspirated /t/, smaller in absolute terms for /d/), and much higher F0 — roughly $1.73\times$ that of male talkers — yet their following vowels are no longer than male talkers’.

The F0 and VOT differences are not on the same footing, however. The F0 gap is very large and physiologically expected: adult female talkers have shorter, less massive vocal folds that vibrate faster, so a higher F0 is essentially guaranteed and functions here as a sanity check that the gender labelling behaves as known, rather than as a discovery about /t/ and /d/. The VOT difference is the more substantive result: it concerns the *timing* of the consonant rather than talker anatomy, and the fact that vowel duration does *not* differ argues against a simple “female talkers lengthen everything” account — if that were the mechanism, the vowels would lengthen too.

Two caveats temper the result: the tests rest on 24 talkers, so the confidence intervals are correspondingly wide; and because talker gender is confounded with any anatomical or speaking-rate differences that covary with it in this sample, the analysis establishes an association rather than a cause. Within

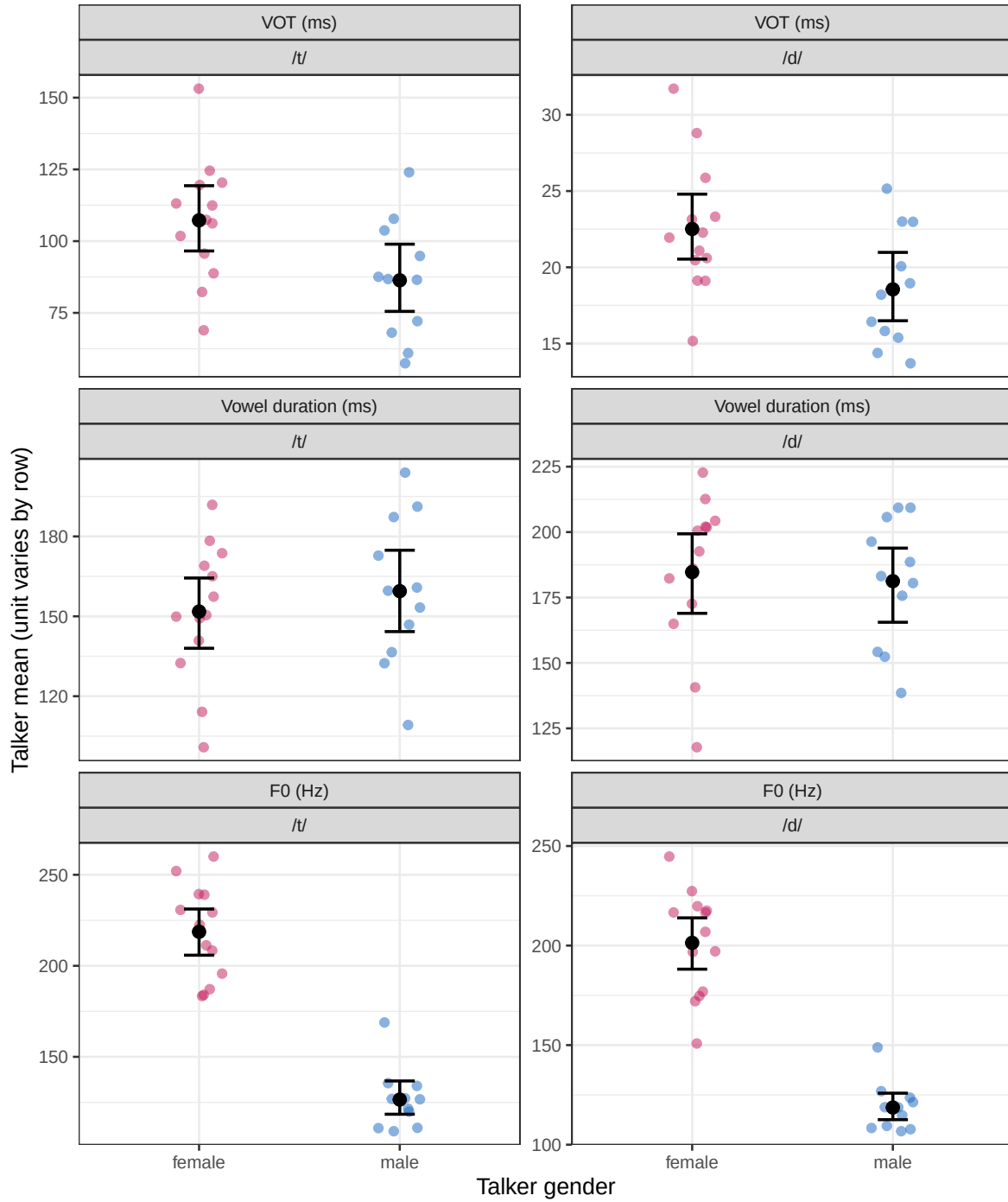


Figure 1: VOT (top, ms), following-vowel duration (middle, ms), and vowel-onset F0 (bottom, Hz) by talker gender, for /t/ (left) and /d/ (right). Each coloured point is one talker's mean; the black point and bars are the group mean and a bootstrapped 95% confidence interval across talkers (mean_cl_boot). Every panel has its own y-axis, and the unit differs by row (see the strip labels). Female and male talkers separate clearly on VOT and F0, but overlap on vowel duration.

those limits, talker gender is a meaningful source of variation in the VOT (and, trivially, the F0) of /t/ and /d/ in the **engCVC** corpus, but not in the duration of the following vowel.